2026 - The Future Of Capitalism

Mock Exam

Instructions

Six questions, one per week of the course. Each question has five sub-parts (a–e) worth 3 points each, for a total of 90 points.

Time allowed: 90 minutes. No books or notes. Answers may be in English or Dutch.

Write concisely. Credit is given for correct reasoning even if expressed differently from the model answer.

Note: provided for reference and exercise. The mock exam should help you get an idea of what kind of questions will be asked, it is not a training program ("if I can do this, I'll get a pass")

Answer Key included at end

Question 1 — Modern Marxism [15 pts]

Based on Lecture 1 and Heilbroner, The Worldly Philosophers, Ch. 6 (pp. 144–148)

a. What is the name of Marx's theory that the material conditions of production shape the laws, norms, and institutions of a society?

Answer:

b. Give one concrete historical example from the lecture that illustrates this theory.

Answer:

c. Marx predicted that capitalism contains the seeds of its own destruction. Name one internal force he identified that drives capitalism toward crisis.

Answer:

d. What concept does Varoufakis use to describe an economy governed by digital platforms rather than market competition?

Answer:

e. Gramsci argued that the ruling class maintains power not only through force but through a subtler mechanism. What is the name of this mechanism, and in one sentence, what does it mean?

Answer:

Question 2 — The Finance View [15 pts]

Based on Lecture 2 and teaching notes 2a & 2b

a. Name two features of the joint-stock corporation (as pioneered by the VOC in 1602) that made it suited to funding risky long-distance ventures.

Answer:

b. Under CAPM, what is the required return on a stock with $\beta = 1.4$, a risk-free rate of 2%, and a market risk premium of 5%?

Answer:

c. In the Diamond-Dybvig model, why can a bank run occur even when the bank is fundamentally solvent?

Answer:

d. A bank has €100 equity, €400 in deposits (four depositors of €100 each), and €500 in loans. If loans must be liquidated at 80 cents on the euro to meet a withdrawal, how much equity does the bank lose per €100 withdrawal? After how many withdrawals is the bank insolvent?

Answer:

e. Name one regulatory tool that addresses the bank-run fragility identified in sub-question c, and explain in one sentence how it works.

Answer:

Question 3 — Economic Nationalism [15 pts]

Based on Lecture 3

a. Country A can produce 100 units of wine or 50 units of cloth. Country B can produce 50 units of wine or 10 units of cloth. Which country has the comparative advantage in cloth? Show your reasoning using opportunity costs.

Answer:

b. Using the same countries: in autarky, A splits production 50/50 (50 wine, 25 cloth) and B splits 50/50 (25 wine, 5 cloth). If A specialises fully in cloth and B fully in wine, what is the new world total of wine and cloth? Is it higher or lower than autarky?

Answer:

c. What is the Easterlin paradox?

Answer:

d. A trade policy produces a gain of €50 for winners and a loss of €30 for losers. Using prospect theory with $\lambda = 2.25$, compute the loss-aversion-weighted net welfare value of this policy. Is it positive or negative?

Answer:

e. René Girard's concept of mimetic desire implies that nations compete for status, not just welfare. Give one example from the lecture of an institution that channels this competitive impulse into cooperation rather than conflict.

Answer:

Question 4 — AI and the Future of Work [15 pts]

Based on Lecture 4, Acemoglu & Restrepo (2019), and Keynes (1930)

a. In the Acemoglu & Restrepo task model, tasks are distributed between [N−1, N]. Capital performs tasks below the automation threshold I; labour performs tasks above I. If N−1 = 0, N = 1.0, and I = 0.4, what is labour's share of tasks?

Answer:

b. Automation raises I from 0.4 to 0.6 (N stays at 1.0). What is the new labour share of tasks? By how many percentage points did it fall?

Answer:

c. New tasks are now created, raising N from 1.0 to 1.2 (I stays at 0.6, N−1 stays at 0). What is the new labour share of tasks? By how many percentage points did it recover compared to sub-question b?

Answer:

d. Frey & Osborne estimated 47% of US jobs at high automation risk; OECD follow-up studies found only 9–10%. In one sentence, what explains this large discrepancy?

Answer:

e. Smith et al. (2019) found that roughly 75% of top pass-through business profits are returns to owner human capital. What happens to those profits when the owner dies or retires?

Answer:

Question 5 — Biodiversity & Inequality [15 pts]

Based on Lecture 5: Weitzman (1998) and the lecture on inequality

a. Weitzman's Noah's Ark model ranks conservation efforts using four factors. Name all four.

Answer:

b. Two species are candidates for conservation. Both have the same cost ($C = 100$). Species A: utility $U = 8$, distinctiveness $D = 10$, $\Delta P = 0.2$. Species B: $U = 10$, $D = 3$, $\Delta P = 0.5$. Compute Weitzman's ranking score ($U \times D \times \Delta P / C$) for each species. Which should be saved first?

Answer:

c. Species B has higher utility and a much larger survival gain (ΔP). In one sentence, why does the model nonetheless rank Species A higher?

Answer:

d. Investor Arjen puts €100 in equities earning 8% per year; saver Paul puts €100 in bonds earning 3% per year. Using the compound interest formula, calculate each person's wealth after 20 years. (You may use $(1.08)^{20} \approx 4.66$ and $(1.03)^{20} \approx 1.81$.)

Answer:

e. Piketty et al. give three causes for rising within-country income inequality since the 1980s. Name all three.

Answer:

Question 6 — Capitalism and the Future of Democracy [15 pts]

Based on Lecture 6: Rodrik (2000), Bikhchandani & Sharma (2001), and Plato

a. Rodrik's trilemma states that hyperglobalization, national sovereignty, and democracy cannot all coexist. Which two did the Bretton Woods era (1945–1971) choose, and which one did it sacrifice?

Answer:

b. In the Bikhchandani & Sharma cascade model: Angela receives signal G and invests. Bob receives signal B but observes Angela's action. What does Bob rationally do, and why?

Answer:

c. Claire also receives signal B and observes that both Angela and Bob invested. What does Claire rationally do, and what does her decision illustrate about information cascades?

Answer:

d. In the cascade model, each signal is correct with probability $p = 0.7$. Before observing any actions, the prior probability that the asset is good is 0.5. After observing one Buy action, what is the posterior probability that the asset is good? Show your calculation using Bayes' rule. (Hint: $P(\text{Buy} \mid \text{good}) = 0.7$, $P(\text{Buy} \mid \text{bad}) = 0.3$.)

Answer:

e. The lecture defines democracy's 'surest value' in a minimal way that does not rely on truth-finding or selecting competent leaders. What is this definition?

Answer:

ANSWER KEY — INSTRUCTOR USE ONLY

Model answers below. Credit correct reasoning even if phrased differently from the model.

Q1 — Modern Marxism

Q1a [3 pts]

MODEL ANSWER:
Historical materialism (also accepted: dialectical materialism, as used by Heilbroner).

Q1b [3 pts]

MODEL ANSWER:
Any one of: the shift from hunter-gatherer to agricultural society; the hand-mill giving feudalism, the steam-mill giving industrial capitalism (Marx's own formulation); the VOC and colonialism restructuring social relations. Credit any historically grounded example tied to the base-superstructure argument.

Q1c [3 pts]

MODEL ANSWER:
Any one of: competition forces wages to subsistence, depressing worker purchasing power and causing crises of overproduction; concentration of capital into fewer hands; the reserve army of labour; the contradiction between socialised production and private ownership. One force, clearly stated, earns full marks.

Q1d [3 pts]

MODEL ANSWER:
Technofeudalism. (Varoufakis: digital platforms extract rent through ownership of cloud capital and network lock-in, not through classical market competition.)

Q1e [3 pts]

MODEL ANSWER:
Cultural hegemony (Gramsci). Meaning: the ruling class maintains dominance by making its worldview appear natural and universal, so the subordinate class consents to the existing order without coercion.

Q2 — The Finance View

Q2a [3 pts]

MODEL ANSWER:

Any two of: limited liability (investors lose only their stake, enabling risk-taking); transferable shares (liquidity, allows pooling capital from dispersed investors); separate legal personality (entity outlives its owners, can hold property and enter contracts); separation of ownership and management (professional management at scale). Two features, each with a brief reason.

Q2b [3 pts]

MODEL ANSWER:
Required return = rf + beta x (rm - rf) = 2% + 1.4 x 5% = 2% + 7% = 9%.

Q2c [3 pts]

MODEL ANSWER:
Banks borrow short (deposits repayable on demand) and lend long (illiquid loans). If many depositors withdraw simultaneously, the bank must liquidate loans at a discount, realising losses that can exceed equity even though the bank would have been solvent had savers waited. The run is self-fulfilling: fear of failure causes the failure.

Q2d [3 pts]

MODEL ANSWER:
Each withdrawal of €100 requires liquidating €100 of loans at 80 cents, yielding €80. The shortfall of €20 is absorbed by equity.

Equity loss per withdrawal: €20.

Starting equity: €100. After 5 withdrawals: 5 x €20 = €100 lost, equity = €0. The bank is insolvent after 5 withdrawals.

Note: there are only 4 depositors, so the bank fails before all depositors have even been served.

Q2e [3 pts]

MODEL ANSWER:
Any one of: deposit insurance (guarantees deposits, removing the incentive to run); capital requirements (equity cushion absorbs losses before insolvency); liquidity requirements (sufficient liquid assets to meet withdrawal demand without fire-sale); central bank as lender of last resort (provides emergency liquidity). One tool, one sentence explanation.

Q3 — Economic Nationalism

Q3a [3 pts]

MODEL ANSWER:

Opportunity cost of cloth:
Country A: producing 1 unit of cloth costs 2 units of wine (100/50 = 2).
Country B: producing 1 unit of cloth costs 5 units of wine (50/10 = 5).

Country A has the lower opportunity cost, so Country A has the comparative advantage in cloth. (Country B has comparative advantage in wine.)

Q3b [3 pts]

MODEL ANSWER:
Autarky totals: wine = 50 + 25 = 75; cloth = 25 + 5 = 30.

With full specialisation: A produces 50 units of cloth (all resources to cloth); B produces 50 units of wine.

New world totals: wine = 50, cloth = 50.

Cloth rises from 30 to 50 (+20 units). Wine stays the same at 50 (B's full capacity). Total output is higher. Both countries can be better off through trade.

Q3c [3 pts]

MODEL ANSWER:
The Easterlin paradox: within countries, richer people report higher happiness than poorer people, but rising average GDP per capita over time does not raise average happiness. Happiness appears to depend on relative income, not absolute income.

Q3d [3 pts]

MODEL ANSWER:
Loss-aversion-weighted welfare = gain - lambda x loss = 50 - 2.25 x 30 = 50 - 67.5 = -17.5.

The result is negative. Despite a positive net monetary value (+20), a loss-averse agent perceives this policy as welfare-reducing because the loss of 30 is weighted 2.25 times more heavily than the gain of 50. This explains why voters may reject policies with positive expected value if the losses are concentrated and salient.

Q3e [3 pts]

MODEL ANSWER:
Any one of: the Olympic Games / Eurovision (competitive desire channelled into rule-governed contests); multinational projects such as Airbus, ESA, or the ISS (nations compete by contributing to a shared goal); Levinas's face-to-face encounter transforming rivalry into relationship. One example with a brief explanation of why it channels rather than inflames competition.

Q4 — AI and the Future of Work

Q4a [3 pts]

MODEL ANSWER:
Labour span = N - I = 1.0 - 0.4 = 0.6. Total task span = N - (N-1) = 1.0 - 0 = 1.0.
Labour share = 0.6 / 1.0 = 60%.

Q4b [3 pts]

MODEL ANSWER:
New labour span = N - I = 1.0 - 0.6 = 0.4. Total span = 1.0.
Labour share = 0.4 / 1.0 = 40%.
Fall: 60% - 40% = 20 percentage points.

Q4c [3 pts]

MODEL ANSWER:
New labour span = N - I = 1.2 - 0.6 = 0.6. Total span = N - (N-1) = 1.2 - 0 = 1.2.
Labour share = 0.6 / 1.2 = 50%.
Recovery: 50% - 40% = 10 percentage points (partial, not full recovery to 60%).

Q4d [3 pts]

MODEL ANSWER:
Frey & Osborne analysed automation risk at the occupation level (can this entire job be automated?), while the OECD used task-level analysis (which specific tasks within a job are automatable?). Even 'at-risk' occupations contain many tasks that resist automation, so the share of jobs predominantly composed of automatable tasks is far smaller.

Q4e [3 pts]

MODEL ANSWER:
Profits fall sharply when the owner dies or retires. They do not simply pass to whoever inherits the financial stake, showing that the profits were tied to the owner's personal skills, networks, and effort rather than to the financial capital alone.

Q5 — Biodiversity & Inequality

Q5a [3 pts]

MODEL ANSWER:

The four factors: U (utility of the species), D (distinctiveness / genetic uniqueness), ΔP (increase in survival probability from the conservation effort), C (cost of the effort).

Q5b [3 pts]

MODEL ANSWER:
Species A: score = (8 x 10 x 0.2) / 100 = 16 / 100 = 0.16.
Species B: score = (10 x 3 x 0.5) / 100 = 15 / 100 = 0.15.

Species A scores higher (0.16 > 0.15) and should be saved first.

Q5c [3 pts]

MODEL ANSWER:
Species A's much higher genetic distinctiveness (D = 10 vs D = 3) contributes more expected diversity to the total gene pool, outweighing Species B's higher utility and larger survival gain.

Q5d [3 pts]

MODEL ANSWER:
Arjen (equities): €100 x (1.08)^20 = €100 x 4.66 = €466.
Paul (bonds): €100 x (1.03)^20 = €100 x 1.81 = €181.

After 20 years, Arjen has €466 and Paul has €181 — a ratio of about 2.6x from identical starting wealth, driven solely by the equity risk premium.

Q5e [3 pts]

MODEL ANSWER:
1. Top marginal tax rates have declined.
2. Workers' bargaining power has declined (labour income share falling).
3. Capital income has increased relative to labour income (the r > g dynamic).

Q6 — Democracy & Information Cascades

Q6a [3 pts]

MODEL ANSWER:
Bretton Woods chose democracy + national sovereignty. It sacrificed hyperglobalization. Capital controls gave governments the policy space to pursue welfare states and full employment; international trade was open but deep financial integration was restricted.

Q6b [3 pts]

MODEL ANSWER:
Bob is indifferent. He infers that Angela got a Good signal (from her Buy action), so the public evidence is 1G vs his own B — exactly balanced. He cannot do better than a coin flip. The lecture assumes he flips a coin; suppose it comes up Invest. He invests not because his signal says so, but because the observable evidence is tied.

Q6c [3 pts]

MODEL ANSWER:
Claire invests, discarding her own private signal. She observes two Buy actions (inferring at least one G signal) against her own single B — the balance of evidence favours investing. Her decision illustrates the key fragility of cascades: once a cascade forms, private information is no longer incorporated into public actions. All subsequent investors rationally ignore their own signals and follow the crowd, even if accumulated private information would point the other way.

Q6d [3 pts]

MODEL ANSWER:
Bayes' rule: P(good | Buy) = P(Buy | good) x P(good) / P(Buy).

P(Buy) = P(Buy | good) x P(good) + P(Buy | bad) x P(bad)
= 0.7 x 0.5 + 0.3 x 0.5 = 0.35 + 0.15 = 0.50.

P(good | Buy) = (0.7 x 0.5) / 0.50 = 0.35 / 0.50 = 0.70.

After observing one Buy action, the posterior probability that the asset is good is 0.70 (up from the prior of 0.50).

Q6e [3 pts]

MODEL ANSWER:
Resolving conflicts without violence — or equivalently, replacing leaders without bloodshed. Democracy's surest value is procedural: it provides a peaceful mechanism for political transition, not a guarantee of wise policy or competent leaders.